

The 9×9 Wrapper Problem

A Self-Contained Statement: Definitions, Gates, Results, and the Resolution

13 September 2026

(gates and negative results, 27 July 2026; resolution, 13 September 2026)

Abstract

Composing the stratified vertical propagator with the disorder-resolved lateral propagator in a single resolvent requires both to speak the same 9-component language. They did not. This note defines every object involved, states every test applied to them and its exact outcome, and then resolves the discrepancy: three defects, two of them inside the stratified 6×6 and one a sign in the source operator. GATE F goes from 0.40–0.81 to 1.1×10^{-15} and GATE D to 5×10^{-16} with no weight at all. The source operator, long suspected, was never the defect: it follows in closed form from the governing equation and agrees with the thesis’s own worked example to 0.000×10^0 . The correction carries over to stratified media unchanged. GATE E turns out to be a *false* gate — the exact reference fails it too, and it was passing because of a defect — which accounts for negative result N1 entirely. What remains is to lattice-sum $\mathbf{P}^z$ into the discrete representation, a construction rather than a search. It is written to be recoverable without reference to any prior discussion: every symbol is defined where it first appears, every numerical claim names the script that produced it.

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1 Conventions and objects

1.1 Indices and ordering

Cartesian indices use the seismological convention of this project:

$$z = 0 \text{ (down)}, \quad x = 1 \text{ (right)}, \quad y = 2 \text{ (out)}, \quad (1)$$

right-handed. Voigt pairs are ordered

$$\alpha = 1, \dots, 6 \longleftrightarrow (p, q) = (0, 0), (1, 1), (2, 2), (1, 2), (0, 2), (0, 1), \quad (2)$$

i.e. (zz, xx, yy, xy, zy, zx) . Shear components carry the *engineering* factor: the strain vector holds 2ε on the last three slots, while the stress vector holds plain σ . This asymmetry is the origin of the weight $\mathbf{W}$ below and is not optional — it is how the existing code is written.

Time convention is $e^{-i\omega t}$ throughout. Units are seismic: km/s, g/cm³, GPa, km.

1.2 The two state vectors

The *displacement–traction 6-vector* is

$$\mathbf{b} = (u_z, u_x, u_y, \tau_{zz}, \tau_{xz}, \tau_{yz})^T, \quad (3)$$

the natural state for a layered medium, since these six quantities are continuous across horizontal interfaces.

The *displacement–strain 9-vector* is

$$\boldsymbol{\psi} = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx})^T, \quad (4)$$

the natural state for a scatterer, since the single-site T -matrix acts on displacement and strain.

1.3 The two weights

$$\mathbf{W} = \text{diag}(1, 1, 1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \quad (9 \times 9), \quad (5)$$

$$\mathbf{J}_6 = \begin{pmatrix} \mathbf{0}_3 & \mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{0}_3 \end{pmatrix} \quad (6 \times 6), \quad (6)$$

the canonical symplectic array of the thesis (its Eq. JDEF). Note $\mathbf{J}_6^T = -\mathbf{J}_6$ and $\mathbf{J}_6^2 = -\mathbf{I}_6$.

2 The three propagators

2.1 $\mathbf{P}^{\text{cf}}$ — the closed-form whole-space propagator

`cubic_scattering.horizontal_greens.exact_propagator_9x9(x, y, z,  $\omega$ , ref)` returns the 9×9

$$\mathbf{P}^{\text{cf}}(\mathbf{r}) = \begin{pmatrix} G & C \\ H & S \end{pmatrix}, \quad (7)$$

built from the Kupradze whole-space Green’s tensor and its derivatives, where G is 3×3 (force $\rightarrow$ displacement), C is 3×6 (Voigt stress source $\rightarrow$ displacement), H is 6×3 (force $\rightarrow$ Voigt strain), and S is 6×6 (Voigt stress source $\rightarrow$ Voigt strain). Its *source* is a pair $(F_i, \Delta\sigma_{V\alpha}^*)$ — a body force and a stress-glut — and its *output* is $\boldsymbol{\psi}$ of (4). This object is treated here as ground truth; §4.1 states the evidence.

2.2 $\mathbf{G}_6$ — the layered displacement–traction Green’s function

`GlobalMatrix.layered_greens.layered_greens_6x6(model,  $\omega$ ,  $k_x$ ,  $k_y$ ,  $j$ ,  $i$ )` returns a 6×6 mapping a source at interface j to the state $\mathbf{b}$ at interface i , at horizontal wavenumber (k_x, k_y) , for a plane-layered model. Two properties matter and are easy to miss:

1. Its **source is a state jump**, not a body force. Internally (`riccati_greens_psv`) a unit discontinuity is injected into the state vector, `rhs_p += I4`.
2. Its **basis is mixed**. The docstring of `riccati_greens_psv` gives it as $[u_x, u_z, \sigma_{zz}/(-i\omega), \sigma_{xz}/(-i\omega)]$: the traction half is scaled by $1/(-i\omega)$ relative to physical traction.

2.3 $\mathbf{A}_{\text{sys}}$ — the Fourier-domain system matrix

The layered problem marches as $\partial_z \mathbf{b} = \mathbf{A}_{\text{sys}} \mathbf{b}$ with (thesis Eq. AKDEF)

$$\mathbf{A}_{\text{sys}}(k_x, k_y) = \begin{pmatrix} 0 & -i\gamma k_x & -i\gamma k_y & a & 0 & 0 \\ -ik_x & 0 & 0 & 0 & b & 0 \\ -ik_y & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -ik_x & -ik_y \\ 0 & -\rho\omega^2 + \zeta k_x^2 + \mu k_y^2 & k_x k_y (\chi + \mu) & -ik_x \gamma & 0 & 0 \\ 0 & k_x k_y (\chi + \mu) & -\rho\omega^2 + \zeta k_y^2 + \mu k_x^2 & -ik_y \gamma & 0 & 0 \end{pmatrix} \quad (8)$$

where, and this is a genuine trap in the thesis,

$$\gamma = \frac{\lambda}{\lambda + 2\mu}, \quad a = \frac{1}{\lambda + 2\mu}, \quad b = \frac{1}{\mu}, \quad \zeta = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad \chi = \frac{2\mu\lambda}{\lambda + 2\mu}, \quad (9)$$

with $\zeta - \chi = 2\mu$. These are *local* to (8); the thesis List of Symbols defines γ, ζ, χ as Chapter 2 contrast ratios, which do *not* apply. A footnote to this effect has been added to the recompiled thesis.

3 The wrapper under test

3.1 Its two operators

`layered_greens_9x9` constructs a 9×9 from $\mathbf{G}_6$ by sandwiching it:

$$\mathbf{M} = \mathbf{A}(\mathbf{k}) \mathbf{G}_6 \mathbf{B}. \quad (10)$$

The **receiver-side operator** $\mathbf{A}$ is `strain_from_displacement_traction`, a 9×6 mapping $\mathbf{b} \mapsto \boldsymbol{\psi}$, i.e. $(u, t) \rightarrow (u, \varepsilon)$:

$$\begin{aligned} \varepsilon_{zz} &= \frac{\sigma_{zz} - \lambda(ik_x u_x + ik_y u_y)}{\lambda + 2\mu}, & \varepsilon_{xx} &= ik_x u_x, & \varepsilon_{yy} &= ik_y u_y, \\ 2\varepsilon_{xy} &= ik_y u_x + ik_x u_y, & 2\varepsilon_{zy} &= \sigma_{yz}/\mu, & 2\varepsilon_{zx} &= \sigma_{xz}/\mu, \end{aligned} \quad (11)$$

with the displacements passed through unchanged. Note $\mathbf{A}$ depends on $\mathbf{k}$ and never sees k_z : it recovers ε_{zz} from σ_{zz} and the in-plane strains.

The **source-side operator** $\mathbf{B}$ is, in the shipped code, `traction_from_strain`: a 6×9 , wavenumber-*independent*, equal to $\begin{bmatrix} \mathbf{I}_3 & 0 \\ 0 & PC \end{bmatrix}$ with C the isotropic Voigt stiffness and P the traction-extraction.

3.2 The corrections applied

Two modifications were derived and applied, giving what is called $\mathbf{M}^{\text{corr}}$ below:

$$\mathbf{M}^{\text{corr}} = \mathbf{A}(\mathbf{k}) [\mathbf{G}_6 \mathbf{Q}] \mathbf{B}^\dagger, \quad \mathbf{Q} = \text{diag}\left(1, -1, -1, \frac{i}{\omega}, -\frac{i}{\omega}, -\frac{i}{\omega}\right), \quad \mathbf{B}^\dagger = -\mathbf{J}_6 \mathbf{A}_{\text{src}}(-\mathbf{k})^T \mathbf{W}. \quad (12)$$

$\mathbf{Q}$ undoes the mixed basis on the source index; $\mathbf{B}^+$ was derived by requiring the swap relation (14) below. $\mathbf{A}_{\text{src}}$ means $\mathbf{A}$ evaluated with the *source* interface’s elastic properties (using the receiver’s leaves an 83% residual).

4 The gates: exact statements and outcomes

Each gate is a necessary condition. Scripts named are in `scripts/` of `MultipleScatteringCalculations`.

4.1 GATE A — the invariant of the closed form

Statement. $\mathbf{W}\mathbf{P}^{\text{cf}}(\mathbf{r})$ is symmetric.

Origin. Measured, not assumed: the elementwise ratio P_{ab}/P_{ba} is an exact outer product w_a/w_b with $w = (1, 1, 1, 1, 1, 1, 2, 2, 2)$ — the Voigt engineering factor of (2).

Outcome. **PASS**, 4.2×10^{-18} at four separations; and 1.0×10^{-16} over twelve further random separations, so it is an identity rather than a coincidence.

4.2 GATE F — the same invariant, for the wrapper

Statement. In the homogeneous limit,

$$\mathbf{W}\mathbf{M}(k_x, k_y) \text{ is symmetric.} \quad (13)$$

Why this follows. $\mathbf{W}\mathbf{P}^{\text{cf}}(x, y, \Delta z)$ is symmetric at every (x, y) ; the 2-D transform over (x, y) is a linear combination with *scalar* coefficients; a linear combination of symmetric matrices is symmetric. Verified numerically as well: after transforming $\mathbf{P}^{\text{cf}}$ on a grid, $\mathbf{W}\tilde{\mathbf{P}}$ is symmetric to 6.4×10^{-16} across twenty wavenumbers.

Restriction. Licensed only in the homogeneous limit, where swapping source and receiver leaves the medium unchanged. For a genuinely stratified reference no within-matrix symmetry should be expected.

Outcome. $\mathbf{M}^{\text{corr}}$ **FAILS**, residual 0.40–0.81 across two frequencies and two slownesses.

4.3 GATE E — the swap relation

Statement.

$$\mathbf{M}(i \leftarrow j)(+\mathbf{k}) = \mathbf{W}^{-1} [\mathbf{M}(j \leftarrow i)(-\mathbf{k})]^T \mathbf{W}. \quad (14)$$

Outcome. $\mathbf{M}^{\text{corr}}$ **PASSES** at 9.7×10^{-16} .

What it cannot see. (14) relates two *different* matrices and is homogeneous of degree one in $\mathbf{M}$. It therefore constrains neither the overall scale (any constant multiplies both sides equally) nor the relation between the G, C, H, S blocks *within* one matrix. Both blind spots proved to be live.

Correction (13 September 2026): GATE E is a *false* gate. It is not merely blind — it is false for the correct propagator, and it was passing *because* of a defect. GATE F gives

$\mathbf{M}_1 = \mathbf{W}^{-1}\mathbf{M}_1^T\mathbf{W}$ and GATE E gives $\mathbf{M}_1 = \mathbf{W}^{-1}\mathbf{M}_2^T\mathbf{W}$; together they force $\mathbf{M}_1 = \mathbf{M}_2$, i.e. $\mathbf{P}(\mathbf{r}) = \mathbf{P}(-\mathbf{r})$. But C and H are *first* derivatives of the Green’s tensor and are therefore *odd* in $\mathbf{r}$, so that can hold only if C vanishes. The two relations differ by exactly a sign on the mixed blocks — which is defect D3. Measured: the *exact* whole-space reference, with no implementation in it at all, fails GATE E by 1.4348×10^{-1} , identical to five figures to the corrected 6×6 , and passes at 7×10^{-17} only when D3 is reintroduced. GATE A is the arbiter, having been measured directly on the validated closed form at twelve random separations. Consequently the exit criterion of `gate_9x9_source_convention.py` (“non-zero unless A, E and F all pass”) can never be met.

4.4 GATE D — the reciprocity law of $\mathbf{G}_6$ itself

Statement. With $\mathbf{SD} = \text{diag}(1, -1, -1, -i\omega, +i\omega, +i\omega)$,

$$\mathbf{G}_6(i \leftarrow j)(+\mathbf{k}) = \frac{i}{\omega} \mathbf{SD} \mathbf{J}_6 [\mathbf{G}_6(j \leftarrow i)(-\mathbf{k})]^T \mathbf{J}_6 \mathbf{SD}. \quad (15)$$

Here $\text{diag}(1, 1, 1, -i\omega, -i\omega, -i\omega)$ is the $\sigma/(-i\omega)$ mixed basis applied uniformly to P–SV and SH, and $\text{diag}(1, -1, -1, 1, -1, -1)$ is the parity of the x, y components under $\mathbf{k} \rightarrow -\mathbf{k}$.

Outcome. **PASS**, 6.3×10^{-16} . Obtained by solving for the weight, not guessing: a joint rank-one fit over interface pairs, slownesses and frequencies has residual 4.8×10^{-15} with $\mathbf{J}_6$ and 0.97 with the identity, so the symplectic signature is required.

Trap. Recovering the SH entries requires $k_y \neq 0$. At $k_y = 0$ the SH channel decouples, components 2 and 5 vanish identically, and a fit returns an unconstrained value for them.

5 Components verified in isolation

Each of these was checked against an independent known answer. None is the defect.

Object	Check	Residual
A , Eq. (11)	Hooke’s law on random plane waves	2×10^{-16}
G₆Q	GATE D, Eq. (15)	6.3×10^{-16}
P^{cf}	GATE A	4.2×10^{-18}
<code>_build_slab_kernels</code>	equals P^{cf} on the lattice	ratio 1.006, 1.001
inverse-FT pipeline	Weyl identity	ratio 0.98–1.03
G₆ solver	translation invariance, homogeneous	$\rightarrow 1.7 \times 10^{-5}$

For **A**: an arbitrary plane wave $u_i = U_i e^{i(k_z z + k_x x + k_y y)}$ gives ε and σ exactly from their definitions; forming **b** and applying (11) reproduces **ψ** to 2×10^{-16} over six random trials, including the non-trivial ε_{zz} row.

For the Weyl check: $\tilde{G} = (i/2k_z) e^{ik_z |\Delta z|}$ transformed by the same pipeline reproduces $e^{ikr}/(4\pi r)$, so the transform normalisation ($\text{ifft} 2 \times N^2 \Delta k^2 / (2\pi)^2$) is correct and no stray 4π exists.

6 The negative results

These close off regions of the search space and are the substantive output.

N1. No $\mathbf{B}$ in the adjoint family satisfies $\mathbf{E}$ and $\mathbf{F}$ together. For $\mathbf{B} \in \{-\mathbf{J}_6 \mathbf{A}_{\text{src}}(-\mathbf{k})^T \mathbf{W}, \mathbf{S} \mathbf{J}_6 \mathbf{A}_{\text{src}}(+\mathbf{k})^T \mathbf{W}, \mathbf{J}_6 \mathbf{A}_{\text{src}}(+\mathbf{k})^T \mathbf{W}\}$ the shipped choice gives $\mathbf{E} = 5.3 \times 10^{-16}$ with $\mathbf{F} = 0.81$, and every variant improving $\mathbf{F}$ destroys $\mathbf{E}$ (best $\mathbf{F} = 0.26$ at $\mathbf{E} = 0.74$). *No $\mathbf{B}$ ever could:* $\mathbf{E}$ and $\mathbf{F}$ are mutually exclusive (§4.3). “Every variant improving $\mathbf{F}$ destroys $\mathbf{E}$ ” was never a clue about $\mathbf{B}$ — it was two incompatible gates pulling in opposite directions. The family also contained the right answer all along (§8).

N2. GATE $\mathbf{F}$ reduces to a property of $\mathbf{G}_6$ alone. If $\mathbf{B} = \mathbf{A}^T \mathbf{W}$ then

$$\mathbf{W} \mathbf{M} = (\mathbf{W} \mathbf{A}) \mathbf{G} (\mathbf{W} \mathbf{A})^T, \quad (16)$$

which is symmetric *if and only if* $\mathbf{G}$ is symmetric. So under that ansatz the question is not about $\mathbf{B}$ at all.

N3. No diagonal source correction makes $\mathbf{G}_6$ symmetric. Requiring $\mathbf{G}_6 \text{diag}(q)$ symmetric gives 15 homogeneous equations in 6 unknowns. The null space does *not* exist: the smallest singular value ratio stalls at $4.9 \times 10^{-3} \rightarrow 1.8 \times 10^{-3}$ over $f = 6 \rightarrow 48$ Hz, the achieved symmetry is only $3\text{--}7 \times 10^{-2}$, and the fitted q drifts with frequency (0.48, 0.53, 0.56, 0.74). The system is inconsistent.

N4. $\mathbf{A}_{\text{sys}}$ is not the conversion needed here. Every grouping $\mathbf{G} \mathbf{Q} \mathbf{A}_{\text{sys}}^{-1}$, $\mathbf{G} \mathbf{Q} \mathbf{A}_{\text{sys}}$, $\mathbf{A}_{\text{sys}}^{-1} \mathbf{G} \mathbf{Q}$ leaves the symmetry residual at 1.4, no better than raw $\mathbf{G}$.

Correction (13 September 2026). The version of this paragraph dated 27 July justified the $\mathbf{A}_{\text{sys}}$ lead by asserting that Box 5.3 of the thesis gives the state-jump $\leftrightarrow$ body-force equivalence as $\mathbf{F} = \mathbf{A}_{\text{sys}} \mathbf{S}$. *It does not.* Box 5.3 assumes a source carrying both a δ and a δ' part, $\partial_z \mathbf{b} = \mathbf{A}_{\text{sys}} \mathbf{b} + F_1 \delta(z - z_s) + F_2 \delta'(z - z_s)$ (the form of Kennett 1983, Eq. 2.78), and gives the *effective* δ coefficient as $F_1 + \mathbf{A}_{\text{sys}} F_2$. So $\mathbf{A}_{\text{sys}}$ multiplies the δ' coefficient, and the object is a source vector, not a conversion applied to a finished Green’s function. Writing $\mathbf{b} = \mathbf{b}_{\text{reg}} + F_2 \delta$ gives $\partial_z \mathbf{b}_{\text{reg}} - \mathbf{A}_{\text{sys}} \mathbf{b}_{\text{reg}} = (F_1 + \mathbf{A}_{\text{sys}} F_2) \delta$: the δ part passes straight through to the jump, and the δ' part enters multiplied by $\mathbf{A}_{\text{sys}}$. The measurement above is sound; the mechanism it refuted was never the thesis’s. The thesis is correct and its Box 5.3 in fact *validates* the operator $\mathbf{B}$ derived in §8 — see (19).

N5. The lateral and stratified propagators are not commensurable as built. In the homogeneous limit the ratio of the stratified spectral kernel to the verified lattice kernel varies by 10–14× across wavenumber and *grows* under damping, so it is not surface leakage. Discretising the stratified propagator onto the lattice improves this 3.8× (spread $9.8 \rightarrow 2.55$) but does not close it; the residual is a roughly constant factor near 12 with $\sim 200\%$ scatter.

7 The question, as it stood

Precisely one question, and it was thought to be small:

Given a 6×6 Green's function whose source index is a *state jump* in the mixed basis of §2.2, what is the correct 6×9 source-side operator $\mathbf{B}$ such that $\mathbf{A}\mathbf{G}_6\mathbf{B}$ reproduces the whole-space $\begin{bmatrix} \mathbf{G} & \mathbf{C} \\ \mathbf{H} & \mathbf{S} \end{bmatrix}$ of (7)?

It was the wrong question. $\mathbf{B}$ was never the defect. N2 was the true signal and it was under-used: *GATE F is a question about $\mathbf{G}_6$.*

8 The resolution

8.1 $\mathbf{B}$, from the governing equation

Write the true traction as $T_{ij} = c_{ijkl}\epsilon_{kl} - m_{ij}$ with m the stress glut and f the body force. Setting $f_i = F_i\delta(z - z_S)$ and $m_{ij} = M_{ij}\delta(z - z_S)$ in the first-order system and integrating across z_S gives the classical source discontinuity vector, $[\mathbf{b}] \equiv \mathbf{b}(z_S^+) - \mathbf{b}(z_S^-)$:

$$\begin{aligned} [u_z] &= aM_{zz}, & [T_{zz}] &= -F_z, \\ [u_x] &= bM_{xz}, & [T_{xz}] &= -F_x + ik_x(M_{xx} - \gamma M_{zz}) + ik_y M_{xy}, \\ [u_y] &= bM_{yz}, & [T_{yz}] &= -F_y + ik_y(M_{yy} - \gamma M_{zz}) + ik_x M_{xy}, \end{aligned} \quad (17)$$

with a, b, γ as in (9). Two features of this kill the search that was actually run: it is *wavenumber-dependent*, and it mixes the glut into *both* halves of the state, so it can never be a diagonal rescaling of anything. In the slot convention the code actually uses (engineering doubling on the shear stress slots), (17) is, entry for entry,

$$\mathbf{B} = \mathbf{J}_6 \mathbf{A}_{\text{src}}(-\mathbf{k})^T \mathbf{W}, \quad (18)$$

which is the shipped $\mathbf{B}^\dagger$ of (12) *up to a global sign* — and both GATE E and GATE F are blind to a global sign. Hence N1.

The thesis validates this exactly. Box 5.3 works a point explosion through its own machinery and obtains the effective source coefficient

$$F_1 + \mathbf{A}_S F_2 = M(\omega) e^{-ik_x z_S} \delta(z - z_S) \left[\frac{1}{\rho \alpha^2}, 0, 0, 0, \frac{2ik_x \beta^2}{\alpha^2}, \frac{2ik_y \beta^2}{\alpha^2} \right]^T. \quad (19)$$

Equation (17) with $M_{ij} = M\delta_{ij}$ gives $[u_z] = aM = M/(\rho\alpha^2)$ and $[T_{xz}] = ik_x M(1 - \gamma) = 2ik_x M\beta^2/\alpha^2$. Relative difference: 0.000×10^0 , on all six components. Two derivations thirty years apart agreeing exactly.

8.2 Three defects

Let $\hat{\mathbf{k}} = (k_x, k_y)/|\mathbf{k}_\parallel|$, $P_{ab}^\parallel = \hat{k}_a \hat{k}_b$ and $P^\perp = \mathbf{I}_2 - P^\parallel$. Define, in the (z, x, y) frame of §2.1, acting on $(\sigma_{zz}, \sigma_{xz}, \sigma_{yz})$,

$$\mathbf{K} = 1 \oplus (-P^\parallel + P^\perp) = 1 \oplus (\mathbf{I}_2 - 2\hat{\mathbf{k}}\hat{\mathbf{k}}^\top), \quad (20)$$

a *Householder reflection* about the plane perpendicular to the horizontal propagation direction: purely geometric, involutive, and independent of the medium and of frequency.

Corrected 14 September 2026. Equation (20) previously carried a factor η_s , the vertical S slowness, on the perpendicular component. That factor was not physics. It compensated a missing η in the layered *SH* eigenvector of the stratified solver, whose traction/displacement ratio was μ where an *SH* wave requires $\mu\eta$. The two errors cancel *exactly* in a uniform medium — there every eigenvector comes from one layer — so every gate in this document passed at 10^{-15} while the *layered SH* reflection was angle-independent, $(\mu_1 - \mu_2)/(\mu_1 + \mu_2)$, instead of the Aki & Richards form $(\mu_1\eta_1 - \mu_2\eta_2)/(\mu_1\eta_1 + \mu_2\eta_2)$. A reflection coefficient that does not depend on the ray parameter is the tell. The two halves are lethal apart: correcting either alone takes the uniform-limit reduction from 10^{-15} to $\sim 3 \times 10^{-1}$. *P* and *SV* were correct throughout. See `scripts/gate_sh_impedance.py`, which holds the uniform limit and the interface reflection simultaneously.

- D1.** `assemble_greens_6x6` rescales the stress source columns on the displacement rows only, so the stress-row $\times$ stress-column block carries *one* factor of $(-i\omega)$ where a consistent change of basis $\mathbf{D}_{\text{row}}\mathbf{G}\mathbf{D}_{\text{col}}$ requires two. Fix: $\mathbf{G}_6[3:6, 3:6] \times (-i\omega)$.
- D2.** The traction half of *both* indices carries $\mathbf{K}$: as $\mathbf{K}$ on the rows, as $(-i\omega)^2\mathbf{K}$ on the source columns, with -1 on the displacement source columns.
- D3.** $\mathbf{B}$ must carry the force and the stress glut with *opposite* relative sign: $\mathbf{B} \rightarrow \mathbf{B} \text{diag}(-\mathbf{I}_3, \mathbf{I}_6)$.

8.3 Evidence

Object	Check	Residual
$\tilde{\mathbf{G}}_{uu}$ reference	inverse transform vs <code>exact_greens</code>	9.2×10^{-6}
traction source columns	vs $\mathbf{K}$, five $(p, \text{azimuth})$	1.8×10^{-14}
moment source columns	$= -\mathbf{I}$ exactly	$\sim 10^{-15}$
traction rows, D1 applied	inside the displacement row space	3×10^{-16}
residual row operator	$= \mathbf{K}$ again	3×10^{-13}
corrected $\mathbf{G}_6$	$=$ exact whole-space jump response	5×10^{-15}
GATE F	was 0.40–0.81	1.1×10^{-15}
GATE D	bare $\mathbf{G}_1 = \mathbf{J}_6 \mathbf{G}_2^T \mathbf{J}_6$, no weight	5.1×10^{-16}
GATE E	false gate; exact ref fails identically	1.4×10^{-1}
stratified, interior planes	slab crossed twice, $Q = 1000$	1.7×10^{-15}

Why the search could not have succeeded. $\mathbf{K}$ carries $K_{xy} = K_{yx} = -2k_x k_y / |\mathbf{k}_\parallel|^2$, two thirds of its weight off-diagonal whenever $k_x k_y \neq 0$. A diagonal ansatz in (z, x, y) is inconsistent by construction — which is exactly N3’s “15 equations, 6 unknowns, no null space”. $\mathbf{K}$ is diagonal

only on the two axes $k_y = 0$ and $k_x = 0$. (In the superseded η_S form the off-diagonal entry was $-(k_x k_y / |\mathbf{k}_\parallel|^2)(1 + \eta_S)$, and the additional slowness dependence made a constant fit hopeless as well.)

The $1/\omega$ residual was a mirage. With $\mathbf{G}_6$ exact, GATE F still failed at a clean $1/\omega$ (0.148, 0.0403, 0.0101 at $f = 12, 48, 192$ Hz). Decomposing the antisymmetric part by block gives 0 on G , 10^{-16} on S , and *exactly* 2.0000 on C and on H — meaning $C = -H^T$. That is a sign, not a $1/\omega$ term. It presents as $1/\omega$ only because $\|C\|, \|H\|$ are $1/\omega$ -sized against $\|S\|$, which dominates the normalising denominator. Searching for a missing $1/\omega$ term in $\mathbf{Q}$ could not have worked.

GATE D, and the thesis. The corrected $\mathbf{G}_6$ obeys $\mathbf{G}(i \leftarrow j)(+\mathbf{k}) = \mathbf{J}_6[\mathbf{G}(j \leftarrow i)(-\mathbf{k})]^T \mathbf{J}_6$ with *no weight whatsoever* — not the $(\mp i\omega)^{\pm 1}$ factors of (15), and not even the parity. This is the law implied by the thesis’s own construction $\mathbf{G} = \mathbf{D}_z(\mathbf{k})\mathbf{Q}^z\mathbf{D}_z^T(-\mathbf{k})\mathbf{J}_6$ together with $\mathbf{D}_z^{-1} = -i\mathbf{J}_6\mathbf{D}_z^T(-\mathbf{k})\mathbf{J}_6$, and it is Wapenaar’s unified $\mathbf{G} = \mathbf{N}\mathbf{G}^T\mathbf{N}$ verbatim. The thesis had the clean law all along; the implementation departed from it.

9 What remains open

The stratified extension — CLOSED. $\mathbf{K}$ needs no stratified generalisation; it is purely geometric. The apparent stratified failure (0.009–0.398 on the marine model) was a *placement* artefact: every interface of that model separates different materials and the test pairs sat *on* them, where the local medium is two-valued and the construction reads a single one per plane. (At the time $\mathbf{K}$ itself was built on the local η_S , which sharpened the same objection; the restriction survives the removal of η_S because the rest of the construction still reads one medium per plane.) With the planes in a layer interior and a fast slab crossed twice between them at $Q = 1000$ — transmission, reflection, conversion and interbed multiples all present — the clean law holds to 1.7×10^{-15} over nine geometries. The control is included: on a discontinuity, no choice of side does better than 3.9×10^{-3} . The restriction costs the application nothing, since scattering voxels live in layer interiors; it is enforced with a diagnostic in `cubic_scattering.layered_correction`.

Upstream or here? — SETTLED by experiment. Applying D1 in-process and running the sibling repository’s own suite gives 119 passed at baseline and 118 passed / 1 failed with the fix. The single failure is not a physical check: it asserts that the 9×9 solver agrees with the 4×4 solver, where the two sides are $(\mathbf{A}\mathbf{G}_6\mathbf{B})$ driven by T_9 and $\mathbf{G}_4$ driven by $\mathbf{B}T_9\mathbf{A}$. Both wrap the *same* raw Green’s function with the *same* $\mathbf{A}$ and $\mathbf{B}$, so it is an algebraic identity that holds for *any* $\mathbf{G}$ and cannot see a common-mode error — it would fail just as readily for a correct change. Verdict: carry the correction here, for a sharper reason than the original one. Not “the error cancels in R and T ”, but *no upstream test compares the 9×9 path to anything outside itself*. Raise a note, not a patch.

N5 — improved, not closed. Re-taken through the corrected wrapper, the kernel-ratio spread roughly halves (pair 2 $\rightarrow$ 0 at $Q = 3$: $34.9 \rightarrow 5.9$) and still does not fall with Q . So the wrapper defects were a real part of N5 but not the whole of it. The remainder is the discrete-versus-continuum aliasing already diagnosed: `_build_slab_kernels` sums the continuum propagator

over discrete voxel positions and transforms, so by Poisson summation its spectral kernel is the continuum kernel *aliased* over reciprocal lattice vectors, while `layered_greens` returns the un-aliased kernel at a single $\mathbf{k}$. The consequence is unchanged and is the next piece of real work: **lattice-sum $\mathbf{P}^z$ into the discrete representation before adding it to $\mathbf{P}^x$** . That is a construction, not a rescaling.

A withdrawn result, recorded. An attempt to confirm the aliasing account by comparing the corrected kernel against a *freshly written* continuum 9×9 showed a uniform 20% offset at every propagating wavenumber, flat in frequency — while the same construction agrees with the *validated* whole-space reference to 5×10^{-15} . The fresh reference was therefore the suspect, and the comparison was withdrawn rather than reported. This is the second time in this work that an unvalidated reference nearly produced a false result; the first was a wavenumber grid that did not contain the S pole ($k_S = \omega/\beta$), which made the absolute reference of §8 look broken at $O(1)$ until the grid was widened. Validate the reference before believing the comparison.

Reproducibility. Gates A–F and the negative results: `scripts/gate_9x9_source_convention.py`; N5: `scripts/composed_matvec.py` and `scripts/discrete_pz_kernel.py`; lateral propagator: `scripts/gate_lateral_sweep_alg52.py`. The resolution: `scripts/gate_displacement_only_calibration.` (the absolute reference and $\mathbf{K}$), `scripts/gate_wrapper_resolution.py` (the three defects, GATE F, GATE D), and `scripts/audit_thesis_ch3_box53.py` ((19) and the Chapter 3 self-consistency checks). A fuller narrative, including the eliminations made and retracted along the way, is in `docs/wrapper_problem_state_2026-09-13.md`.